Synthetic data. Prototype built by Russell Harris. Not affiliated with or endorsed by CarbonAI. Not a CarbonAI product and does not describe how CarbonAI works.

Covenant Desk: a quarterly loan review where every covenant figure traces to the loan agreement

Template 1 quarterly surveillance report as rendered in the base state: both Pellham readings carried, no support yet for the February 2026 reclass. Every figure comes from the figure registry; the web version links each to its agreement section or document line.

Linden Hollow Plaza (synthetic). Quarterly surveillance report, Test Date 2026-09-30. Synthetic data.

Tallis Reach Credit (synthetic)

LOAN

Tallis Reach Credit, original $14,000,000.00, balance $13,959,940.85 at the Test Date, 6.25% fixed, amortizing from 2026-07-01, maturity 2029-07-01. Package received 2026-10-27, due 2026-11-14.

COVENANT TESTS

Borrower certificate: DSCR 1.64x, debt yield 10.72%, no trigger reported.

Lender NOI under the loan agreement, R1, Pellham deferral approved in writing: $1,319,198.25; Debt Service $1,034,404.92; DSCR 1.28x against a 1.25x minimum; debt yield 9.45% against a 9.75% trigger.

Lender NOI under the loan agreement, R2, Pellham no approved deferral: $1,285,005.75; Debt Service $1,034,404.92; DSCR 1.24x against a 1.25x minimum; debt yield 9.20% against a 9.75% trigger.

Needs info: DSCR covenant result depends on the Pellham deferral; obtain the signed deferral agreement and Lender's written consent under Section 5.9, or servicing's confirmation that none exists. Cash Management Trigger Event in effect under every reading.

> Lender NOI is $1,319,198.25 under R1 and $1,285,005.75 under R2, against Debt Service of $1,034,404.92. DSCR is 1.28x under R1 and 1.24x under R2, versus the 1.25x minimum, and debt yield is 9.45% under R1 and 9.20% under R2, versus the 9.75% trigger. Review result is needs info because the DSCR covenant outcome depends on the Pellham signed deferral agreement and Lender's written consent under Section 5.9, or servicing's confirmation that none exists, and the Cash Management Trigger Event is in effect under every reading. Proposed rating is 6 from 5, subject to credit officer approval.
> Model draft (gpt-5.4-mini-2026-03-17), passed the numbers gate on retry

CERTIFICATE TO LENDER BRIDGE

Termination fee excluded -$54,000.00; Pellham exclusion, R1, deferral approved: $0.00; management fee at the floor, R1, deferral approved: -$20,962.25; reserves -$84,000.00; February 2026 repairs restatement -$18,400.00. Debt Service after interest-only burn-off $119,553.66.

Termination fee excluded -$54,000.00; Pellham exclusion, R2, no approved deferral: -$35,250.00; management fee at the floor, R2, no approved deferral: -$19,904.75; reserves -$84,000.00; February 2026 repairs restatement -$18,400.00. Debt Service after interest-only burn-off $119,553.66.

> The certificate reports NOI of $1,496,560.50, debt service paid of $914,851.26, DSCR of 1.64x, and debt yield of 10.72%. Under the loan definition, NOI is reduced by the lease termination fee, reserves, the management fee floor, and the open prior-period restatement, while debt service is measured as the greater of paid debt service and scheduled principal and interest under Section 1.1, which moves lender NOI to $1,319,198.25 and DSCR to 1.28x.
> Model draft (gpt-5.4-mini-2026-03-17), passed the numbers gate

PELLHAM FITNESS

Pellham Fitness: rent roll Current; AR aging $35,250.00 unpaid, $11,750.00 over 90 days; the borrower letter claims a deferral agreed with servicing; consent log empty. DSCR 1.28x with the deferral approved and 1.24x without it. Decision-relevant: waits on the signed deferral agreement and Lender's written consent under Section 5.9, or servicing's confirmation that none exists.

> Pellham Fitness (6,000 SF) is a Major Lease because it exceeds 5,000 SF, so Section 5.9 requires Lender's prior written consent for any rent deferral. The rent roll shows it Current, the AR aging shows $35,250.00 unpaid with $11,750.00 over 90 days, and the borrower letter says a deferral was agreed with servicing while the consent log is empty and no deferral agreement is in the package. Both readings are carried, with DSCR 1.28x under one reading and 1.24x under the other, and the signed deferral agreement and Lender's written consent are needed to settle the result; absent those, the deferral does change the covenant result.
> Model draft (gpt-5.4-mini-2026-03-17), passed the numbers gate

PRIOR QUARTER AND SETUP

116 of 117 overlapping T-12 cells agree; February 2026 repairs restated from $26,900.00 to $8,500.00.

Prior review NOI $1,342,114.50 reproduced as $1,342,114.50. The servicing record carries a 9.00% trigger from the term sheet; the agreement says 9.75%. At the prior Test Date debt yield was 9.59%, so the trigger was met then.

GUARANTOR

Liquid Assets $1,525,000 under the agreement definition (PFS total $1,980,000 includes retirement accounts) against $1,500,000, met; net worth $18,940,000 against $15,000,000, met.

WATCH LIST

- W1 covenant DSCR below 1.35x: every reading.
- W2 Cash Management Trigger Event in effect: every reading.
- W3 Pellham Fitness, 6.63% of base rent, on an unapproved deferral or past due: every reading.
- W4 February 2026 repairs restated without support ($18,400.00): every reading.
- W5 guarantor liquidity headroom $25,000, within 10% of the minimum: every reading.
- W6 $118,125.00 of Test Period revenue from a suite now vacant: every reading.

Forward in-place DSCR (policy view, not a covenant test), R1, deferral approved: 1.16x.

Forward in-place DSCR (policy view, not a covenant test), R2, no approved deferral: 1.13x.

PROPOSED RATING

Proposed 6 (Special Mention) from 5 under every reading: worse of the covenant DSCR band and the forward in-place DSCR band (P6). Reviewer: approve or override with a reason.

ACTIONS

Hold under every reading:

- A1 Send the Cash Management Trigger Event notice under Section 6.13: debt yield 9.45% and 9.20%, below 9.75%.
- A2 Propose rating 6 (Special Mention) from 5, for the credit officer to approve.
- A3 Correct the servicing setup: cash management trigger to the agreement value (9.75%, not 9.00% from the term sheet).
- A4 Request invoice or contract support for the February 2026 reclass of repairs to capital ($18,400.00); P3 applied until then.
- A5 Add to the watch list: W1, W2, W3, W4, W5, W6.

Wait on the signed deferral agreement and Lender's written consent under Section 5.9, or servicing's confirmation that none exists:

- A6 Send the covenant failure notice and cure demand of $87,000 under Section 6.14 if no approved deferral exists.

Reviewer: approve each action.

What this is

A post-close surveillance workflow on one synthetic retail loan: a loan agreement excerpt, the servicing record and loan setup, the borrower's compliance certificate, a monthly T-12 with GL notes, a rent roll exported with continuation charge lines and subtotals, an AR aging, the borrower's cover letter, the lender's own prior-quarter review, a guarantor PFS and a written surveillance policy. Rules recompute the certificate under the agreement's definitions (termination fee exclusion, management fee floor, reserves, Debt Service after the interest-only burn-off, delinquent-tenant exclusion), reconcile the package against the prior review and the servicing setup, test the guarantor, apply watch triggers and propose a rating. Where documents disagree the desk carries every reading; when the readings give different covenant results it returns Needs info, names the document that settles it, and separates the actions that hold under every reading from the one that waits.

How it is checked

An answer key written when the issues were planted, an independent Python recomputation with Decimal and half-up rounding, a lineage check from every memo figure to an agreement section, a document line or a policy rule, an anchor-text check that each cited line shows the value, and a release gate that blocks simulated regressions. Results on this build:

- data.json schema: 0 issues
- 7 of 7 parsed control totals equal the printed totals and the key (rent roll SF, BASE, REC; AR; both T-12s)
- 4 of 4 definitions (D1 to D4) and 5 of 5 findings (F1 to F5) with exact type, outcome and amount
- 0 findings outside the answer key
- 4 of 4 decoys (DC1 to DC4) produce no finding and appear in Checked and agreed
- 12 of 12 bridges foot to the cent (certificate NOI to lender NOI per reading, and certificate debt service to Debt Service, five states)
- 9 of 9 review states match the key (base, reclass support, R1 adopted, R2 adopted, cure $86,000 and $87,000, FX1 to FX3)
- Loan setup: 14 terms compared, 1 mismatch (cash management trigger); prior Test Date retest: met under the agreement, not met under the servicing value
- Prior quarter: 117 cells compared, 116 agree, 1 restated; prior NOI reproduced to the cent from its own T-12
- Guarantor: Liquid Assets $1,525,000 under the agreement definition, headroom $25,000; both tests met; W5 fires
- Baselines B0 (certificate), B1 (servicing-style check) and B2 (prior rating carried forward) reproduce: compliant, no trigger, rating carried
- 787 of 787 figures and results match the independent Python implementation (five states)
- 263 of 263 rendered memo figures in both templates resolve to source, definition or policy anchors (base, reclass, R2 adopted)
- 101 of 101 source, definition and policy anchors: the cited line shows the figure's value
- Template switch: 164 figure values identical across both templates and equal to the registry (455 renders)
- Orphan-number scan: 0 memo renders with a number outside a figure placeholder; negative control caught 2 of 2 typed numbers
- 8 of 8 gate tests return the expected answer
- 9 of 9 model paragraphs pass the gate (0 replaced after the gate); 2 of those rejected on hand review and replaced by rule drafts; 9 of 9 paragraphs shown pass; 13 API calls (cap 30)
- 5 of 5 simulated regressions blocked: certificate figures trusted, Debt Service on actual payments, Pellham resolved from rent roll status, trigger from the servicing setup, PFS stated liquidity.
- Narrative paragraphs drafted by gpt-5.4-mini-2026-03-17 from a facts object, each passed through the numbers gate and reread by hand (2 of 9 rejected on the reread and shown as rule drafts): 13 API calls, estimated cost $0.02 at an assumed list price. The model cannot set a number, a result, the rating or an action.

Where the model and the reviewer sit

Rules apply the definitions and compute every figure. The model writes three paragraphs and may only write a number that equals the figure it tags; the gate also blocks any form of "default", recommendation language, a rating change or notice described as done, and any mention of Pellham that does not give both DSCRs and name the deferral agreement and written consent. The credit officer approves or overrides the rating with a reason, approves each action, and may adopt a reading or accept the reclass support with a logged reason. The evals run on the default state and never read reviewer actions.

Limits

- Documents are authored text pages, not PDFs, scans or servicing system exports; extraction is mocked. The loan agreement is pre-abstracted.
- The surveillance policy, rating bands and both templates are authored for this demo and do not describe any real lender, including the author's employer.
- The same person planted the issues and wrote the answer key. The results show the logic, the lineage and the deferral behavior on a known case. They do not show accuracy on real loans, adoption by a credit team, or time saved.
- The forward in-place view and the rating rule are written policy choices, not covenant tests.

Notice

Every lender, property, borrower, guarantor, tenant, document and number in this brief is invented. Prototype built by Russell Harris as a work sample. Not affiliated with or endorsed by CarbonAI, not a CarbonAI product, and it does not describe how CarbonAI works. Interactive version: russh.work/covenant-desk